

Lecture 27 : Private-key cryptography (Part I)

Introduction

In our increasingly interconnected world, the security of communication is paramount. Whether it's personal messages, financial transactions, or sensitive government data, ensuring that information remains confidential and authentic is a fundamental challenge. Encryption and decryption stand as cornerstones of this security, transforming readable data into an unreadable format and back again, thereby protecting it from unauthorized access. This lecture delves into the foundational concepts of private-key cryptography, also known as symmetric-key cryptography, which forms the basis for many secure communication systems.

This first part of our discussion on private-key cryptography will focus on understanding its core principles and exploring some classical techniques. While these historical methods may not be robust enough for modern security needs, they provide an invaluable conceptual framework for grasping how encryption works. We will examine how a shared secret key enables secure communication between parties and highlight the inherent strengths and weaknesses of these early cryptographic algorithms, setting the stage for understanding the more sophisticated practical algorithms used today.

Core Concepts

Term	Definition	Significance
Plaintext	The original, readable message or data that needs to be protected.	This is the sensitive information that cryptography aims to secure from unauthorized access.
Ciphertext	The garbled, unreadable form of the plaintext after it has been encrypted.	This is the protected version of the message transmitted over insecure channels; an intruder should not be able to understand it.
Key	A secret value, typically a string of bits or characters,	The security of symmetric-key cryptography

Term	Definition	Significance
	used in conjunction with an algorithm to encrypt and decrypt data.	fundamentally relies on the secrecy and proper management of this shared value.
Encryption Algorithm (EA)	A mathematical function that takes plaintext and a key as input to produce ciphertext.	This is the process that transforms readable data into its secure, unreadable form.
Decryption Algorithm (DA)	A mathematical function that takes ciphertext and the same key as input to recover the original plaintext.	This is the process that reverses encryption, allowing the intended recipient to read the original message.
Symmetric-key Cryptography	A cryptographic system where the same secret key is used by both the sender for encryption and the receiver for decryption.	It simplifies key management for two parties but becomes complex for many, and its security depends entirely on key secrecy.
Substitution Cipher	A type of cipher where characters or groups of characters in the plaintext are systematically replaced by other characters or groups.	Represents a fundamental approach to encryption, altering the content of the message rather than its structure.
Transposition Cipher	A type of cipher where the order of characters in the plaintext is rearranged or permuted, but the characters themselves remain unchanged.	Represents another fundamental approach, altering the structure of the message rather than its content.
Stream Cipher	An encryption method that encrypts and decrypts data continuously, bit by bit or byte by byte, as it flows.	Suitable for real-time communication where data arrives sequentially, like streaming video or audio.

Term	Definition	Significance
Block Cipher	An encryption method that processes data in fixed-size blocks, encrypting or decrypting an entire block at once.	Offers high security and is widely used for encrypting stored data or large files, often requiring padding for incomplete blocks.

Detailed Analysis

1. Fundamentals of Symmetric Key Cryptography

Symmetric-key cryptography, also known as private-key cryptography, operates on a straightforward principle: a single, secret key is shared between the sender and the receiver. This shared secret is crucial for both encrypting the original message (plaintext) into an unreadable form (ciphertext) and for decrypting the ciphertext back into its original plaintext.

The process can be functionally expressed as:

* **Encryption:** $C = E(P, K)$ where C is the ciphertext, P is the plaintext, K is the shared secret key, and E is the encryption algorithm.

* **Decryption:** $P = D(C, K)$ where P is the recovered plaintext, C is the ciphertext, K is the same shared secret key, and D is the decryption algorithm.

The term "symmetric" highlights that the same key is used for both operations, while "private" emphasizes that this key must remain secret between the communicating parties. A fundamental security principle in this context is that the overall security of the scheme should not depend on the secrecy of the algorithm itself. Instead, the security must solely rely on the secrecy of the key. This means that even if an intruder knows the encryption and decryption algorithms, they cannot decode the ciphertext without possessing the correct key.

However, a significant challenge with symmetric-key cryptography arises in managing keys, especially in a network with many communicating parties. If n parties need to communicate securely with each other, each pair requires a unique shared secret key. The total number of distinct keys required for n parties is given by the combination formula $n C 2$, which simplifies to $n * (n - 1) / 2$. For instance, if $n = 1000$, the number of keys would be approximately 500,000. Maintaining the secrecy of such a large number of keys becomes a complex and difficult task, posing a practical limitation for large-scale symmetric-key systems.

2. Classical Private Key Algorithms - Substitution Ciphers

Classical private-key algorithms primarily dealt with text messages and can be broadly categorized into substitution and transposition ciphers. Substitution ciphers work by replacing letters or groups of letters in the plaintext with different ones.

Caesar Cipher

One of the oldest and simplest substitution ciphers is the Caesar Cipher. It involves replacing each letter in the plaintext with the letter that appears a fixed number of positions (e.g., three) down the alphabet. For example, if the shift is 3, 'A' becomes 'D', 'B' becomes 'E', and so on. The alphabet is treated cyclically, meaning 'Z' would wrap around to 'C' (Z → A → B → C).

The generalized Caesar cipher introduces a key k , which is the number of positions to shift. If we assign numerical values to alphabets (A=1, B=2, ..., Z=26), the encryption and decryption formulas are:

* **Encryption:** $C = (P + k - 1) \% 26 + 1$

* Example: Encrypt 'B' (P=2) with k=5.

$C = (2 + 5 - 1) \% 26 + 1 = 6 \% 26 + 1 = 6 + 1 = 7$. The 7th letter is 'G'. So, 'B' becomes 'G'.

* Example: Encrypt 'Y' (P=25) with k=5.

$C = (25 + 5 - 1) \% 26 + 1 = 29 \% 26 + 1 = 3 + 1 = 4$. The 4th letter is 'D'. So, 'Y' becomes 'D'.

* **Decryption:** $P = (C - k + 25) \% 26 + 1$ (adjusting for negative results by adding 25 or 26 before modulo)

The Caesar cipher is extremely vulnerable to brute-force attacks because the key space is very small (only 25 possible shifts, as a shift of 26 is equivalent to 0). An attacker can simply try all possible key values until a meaningful plaintext is recovered.

Mono-alphabetic Cipher

To enhance security beyond the Caesar cipher, the Mono-alphabetic cipher allows for an arbitrary substitution of letters. Instead of a fixed shift, each letter of the alphabet is mapped to a unique, different letter. For example, 'A' might map to 'P', 'B' to 'Z', 'C' to 'A', and so on. The key for this cipher is the specific permutation of the 26 alphabets.

The number of possible keys for a mono-alphabetic cipher is 26 factorial (26!), which is approximately 4×10^{26} . This vast key space makes brute-force attacks impractical. However, this cipher is still easily breakable using frequency analysis. In any natural language (like English), certain letters (e.g., 'E', 'T', 'A') and letter pairs

(digrams like 'TH', 'HE') or triplets (trigrams) occur with predictable frequencies. By analyzing the frequency of letters, digrams, and trigrams in the ciphertext, an attacker can deduce the most likely plaintext mappings and reconstruct the original message.

3. Classical Private Key Algorithms - Transposition Ciphers

Transposition ciphers, unlike substitution ciphers, do not change the letters themselves but rather rearrange their order. The letters are permuted to obscure the original message.

A common method involves writing the plaintext into a rectangle, row by row, and then reading it out column by column in a specific, key-defined order.

- * **Key:** The key consists of two parts: the number of columns in the rectangle and the order in which these columns are read.

- * **Encryption Example:**

- * Plaintext: "WELCOME TO THE NPTEL COURSE ON ETHICAL HACKING"

- * Assume a key specifies 7 columns. The plaintext is written row-wise, filling spaces with dashes:

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W E L C O M E
T O T H E N P
T E L C O U R
S E O N E T H
I C A L H A C
K I N G - - -
```

- * If the key specifies reading columns in the order 1, 2, 3, 4, 5, 6, 7, the ciphertext would be formed by concatenating the letters from column 1, then column 2, and so on. If the key specifies a different order (e.g., 4th column, then 2nd, then 1st, etc.), the ciphertext would be even more jumbled.

- * For example, reading column 1 first: W T T S I K

- * Then column 2: E O E E C I

- * And so on, until all columns are read in the specified order.

Transposition ciphers also suffer from vulnerabilities. Since the letters themselves are not changed, the letter frequencies in the ciphertext remain identical to those in the original plaintext. An attacker can guess the number of columns (a relatively small number of possibilities) and then look for common words or patterns. If a probable word (e.g., "ethical") is known to be in the message, an attacker can look for its letters appearing with gaps corresponding to the number of columns, and then deduce the column order. This makes them relatively easy to break, especially with computational assistance.

4. Practical vs. Classical Ciphers & Data Types

The classical ciphers discussed (Caesar, Mono-alphabetic, Transposition) are primarily conceptual and historically significant. They are simple, often designed for manual encryption, and limited to text-based messages. They are not considered practical for modern security needs due to their inherent weaknesses and ease of breaking.

Practical ciphers, in contrast, are far more complex and are designed to be implemented by computers. They can encrypt any type of digital data, not just text. This includes binary files, images, audio clips, and videos. The complexity of these algorithms makes manual encryption or decryption virtually impossible, requiring computational power for their operation. These advanced algorithms are the backbone of contemporary secure communication.

5. Stream vs. Block Ciphers

Modern practical ciphers can be broadly classified into two categories based on how they process data:

- **Stream Ciphers:** These ciphers encrypt and decrypt data bit by bit or byte by byte as it arrives in a continuous stream. They are ideal for real-time communication where data is constantly flowing, such as streaming video, audio, or secure voice calls. The encryption process generates a keystream that is combined with the plaintext stream to produce the ciphertext stream.
- **Block Ciphers:** These ciphers operate on fixed-size blocks of data. A block cipher takes an n -bit block of plaintext and, using the key, produces an n -bit block of ciphertext. Common block sizes include 64, 128, or 256 bits. If the plaintext message length is not an exact multiple of the block size, padding (adding extra bits, often zeros or ones) is applied to the last block to ensure it reaches the required length. Block ciphers are widely used for encrypting stored data, files, and network protocols where data can be processed in discrete chunks.

Key Takeaways

- [] Symmetric-key cryptography uses a single, shared secret key for both encryption and decryption.
- [] The security of symmetric-key schemes relies on the secrecy of the key, not the algorithm.
- [] Key management is a significant challenge for symmetric-key systems, especially with many communicating parties ($N(N-1)/2$ keys).

- [] Classical substitution ciphers (like Caesar and Mono-alphabetic) replace letters but are vulnerable to brute-force or frequency analysis.
- [] Classical transposition ciphers rearrange letter order but maintain original letter frequencies, making them susceptible to pattern analysis.
- [] Classical ciphers are not practical for modern security due to their simplicity and ease of breaking.
- [] Modern practical ciphers are complex, computer-dependent, and can encrypt any type of digital data.
- [] Stream ciphers encrypt data bit by bit for continuous data flows.
- [] Block ciphers encrypt data in fixed-size blocks, often requiring padding.

Conclusion

This lecture has provided a foundational understanding of private-key cryptography, tracing its evolution from simple, classical techniques to the conceptual underpinnings of modern systems. We've seen how the core idea of a shared secret key enables secure communication, transforming plaintext into garbled ciphertext and back again. While historical methods like the Caesar and Mono-alphabetic substitution ciphers, and transposition ciphers, offered early attempts at secrecy, their inherent vulnerabilities to attacks like brute force and frequency analysis underscore the need for more robust solutions.

The transition from these classical, text-based methods to complex, computer-driven algorithms capable of handling any data type marks a significant leap in cryptographic capabilities. The distinction between stream and block ciphers further illustrates the diverse approaches to securing data, whether it's a continuous flow or discrete chunks. This conceptual groundwork is essential for appreciating the sophistication of contemporary cryptographic algorithms. In the next lecture, we will build upon these principles by exploring the practical, widely-used private-key algorithms that form the bedrock of today's secure digital world.